



Product Requirement Document (PRD)



Cab Services Billing & Vendor Management System

1. Product Overview

A centralized software platform designed for cab service companies serving IT clients to:

- Track trips (vehicle-wise, vendor-wise, client-wise)
 - Automate invoice generation for clients
 - Manage vendor payments and settlements
 - Provide real-time analytics (KM, trips, cost, margins)
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2. Objectives

- Reduce manual billing effort by **80–90%**
 - Eliminate errors in trip calculations and invoices
 - Speed up vendor payouts
 - Provide transparency to clients and vendors
 - Improve profitability tracking
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3. Target Users

1. **Admin (Cab Company Owner / Ops Manager)**
 2. **Accounts Team**
 3. **Vendors (Cab Owners / Drivers)**
 4. **Corporate Clients (IT Companies)**
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4. Problem Statement

Current challenges faced by cab service providers:

- Manual tracking of trips in Excel
 - Delays in invoice generation
 - Errors in KM calculations and pricing
 - Difficulty in vendor payment reconciliation
 - No centralized dashboard for performance tracking
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5. Solution Overview

A cloud-based platform that:

- Captures trip data (manual or API/GPS integration)
 - Auto-calculates billing based on predefined pricing models
 - Generates invoices instantly
 - Tracks vendor dues and automates settlements
 - Provides analytics dashboard
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6. Core Features

6.1 Trip Management System

- Add/Edit Trips
 - Fields:
 - Date & Time
 - Pickup & Drop Location
 - Client Name
 - Vendor Name
 - Vehicle Type
 - KM traveled
 - Waiting time
 - Toll/Parking
 - Bulk Upload via Excel
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6.2 Pricing Engine

Custom pricing rules per client:

- Per KM rate
- Per hour rate

- Minimum guarantee (e.g., 40 KM/day)
 - Night charges
 - Waiting charges
 - Surge pricing
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6.3 Client Billing (Invoice Generation)

- Auto-generate invoices (daily/weekly/monthly)
 - GST calculation
 - Invoice PDF download & email
 - Client-wise billing summary
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6.4 Vendor Payment Module

- Track vendor earnings per trip
 - Deduct commission
 - Generate vendor payout reports
 - Payment status (Paid / Pending)
 - Vendor invoice generation
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6.5 Dashboard & Analytics

- Total trips
 - Total KM
 - Revenue vs Cost
 - Profit margins
 - Vendor performance
 - Client-wise revenue
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6.6 Notifications & Automation

- Auto email invoices to clients
 - WhatsApp notifications (optional)
 - Payment reminders
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6.7 Role-Based Access

- Admin (Full Access)
 - Accounts (Billing + Payments)
 - Vendor (Limited access to trips & earnings)
 - Client (View invoices only)
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6.8 Reports Module

- Trip Report
 - Client Invoice Report
 - Vendor Payment Report
 - Profit & Loss Report
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7. Advanced Features (Differentiators)

- GPS integration for auto KM tracking
 - Mobile app for drivers to update trips
 - API integration with corporate systems
 - Dynamic pricing engine
 - AI-based cost optimization (future scope)
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8. Workflow

Step 1: Trip Entry

Trip data entered manually or via API

Step 2: Pricing Applied

System calculates cost based on rules

Step 3: Invoice Generated

Client invoice created automatically

Step 4: Vendor Settlement

Vendor payout calculated

Step 5: Reports Generated

Dashboard updated in real-time

9. 🧱 Tech Requirements

Frontend:

- Web dashboard (React / Angular)

Backend:

- Node.js / Python

Database:

- PostgreSQL / MongoDB

Integrations:

- Email (SMTP)
 - WhatsApp API
 - Payment Gateway
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10. 📱 MVP Scope (Phase 1)

Focus on:

- Trip entry
 - Pricing rules
 - Invoice generation
 - Vendor payment tracking
 - Basic dashboard
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11. 🚀 Future Scope

- Mobile app for drivers
 - AI-based route optimization
 - Predictive demand analysis
 - Integration with HR systems of IT companies
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12. 💰 Revenue Model

If you plan SaaS:

- Subscription model:
 - ₹2,999/month (Basic)
 - ₹5,999/month (Pro)
 - ₹9,999/month (Enterprise)

OR

- Per trip billing (₹1–₹3 per trip)
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13. ⚠️ Risks & Challenges

- Incorrect trip data entry
 - Complex pricing rules per client
 - Resistance to shift from Excel
 - Vendor adoption challenges
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14. ✅ Success Metrics

- Time to generate invoice ↓
 - Billing errors ↓
 - Vendor payout cycle time ↓
 - Monthly recurring revenue ↑
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15. 🔥 Extra Features You SHOULD Add (Game-Changer)

- Auto KM verification via Google Maps API

- Client login portal (self-service invoices)
- Vendor mobile app for trip updates
- Smart alerts for unusual billing patterns
- Margin tracker per trip (profit visibility)

CabMitra

Product Requirements Document (PRD)

Cab Services Billing & Vendor Settlement Management Platform

Version: 1.0

Product Type: SaaS / Transport Operations ERP

Prepared For: Product, Design, Engineering, Operations, Stakeholders

1. Product Overview

Product Name

Cab Operations & Vendor Settlement Platform

Product Summary

A centralized transport operations platform designed for cab service providers serving corporate/IT clients.

The platform automates:

- multi-format trip data processing
- pricing calculations
- vendor settlements
- invoice generation
- operational analytics

The system converts heterogeneous client transport data into a standardized operational and financial workflow.

2. Product Vision

To become the operating system for corporate employee transport vendors by replacing fragmented Excel-based workflows with a scalable, automated transport ERP platform.

3. Problem Statement

Corporate transport vendors currently manage operations using:

- multiple Excel sheets
- WhatsApp coordination
- manual calculations
- disconnected billing systems

This creates:

- billing errors
- delayed invoices
- vendor disputes
- poor visibility into profitability
- reconciliation issues
- operational inefficiency

Additionally:
every corporate client shares trip data in different formats, making manual normalization extremely time-consuming.

4. Product Goals

Primary Goals

G1 — Automate Billing Workflow

Reduce manual billing effort by 80–90%.

G2 — Standardize Multi-Client Data

Normalize different Excel formats into one operational system.

G3 — Automate Vendor Settlements

Reduce payout calculation time and disputes.

G4 — Improve Profit Visibility

Enable real-time margin tracking per trip/vendor/client.

G5 — Replace Excel Operations

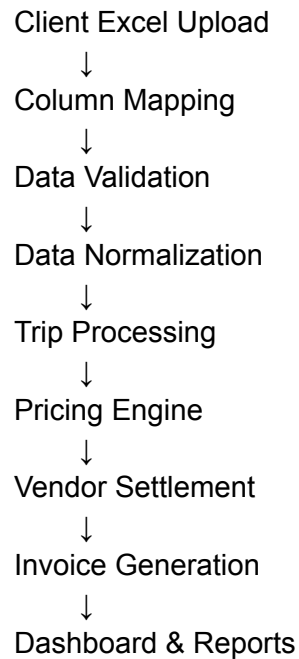
Create a centralized operational workflow platform.

5. Target Users

User Type	Responsibility
Admin	Full platform management
Operations Team	Upload & validate trips
Accounts Team	Billing & settlements

Vendors	View invoices & payouts
Corporate Clients	View billing reports (future scope)

6. Core Business Workflow



7. Product Scope

Included in MVP

- ✓ Excel Upload
- ✓ Dynamic Header Mapping
- ✓ Trip Management
- ✓ Pricing Engine
- ✓ Settlement Engine
- ✓ Invoice Generation
- ✓ Dashboard & Reports

- ✓ Vendor Settlement Reports
 - ✓ Role-Based Access
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Excluded from MVP

- ✗ GPS Tracking
 - ✗ Driver Mobile App
 - ✗ Route Optimization
 - ✗ AI Analytics
 - ✗ Corporate Employee Portal
 - ✗ Automated Dispatching
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8. Key Product Modules

8.1 Authentication & User Management

Features

- Secure Login
- Role-Based Access
- User Permissions

Roles

- Admin
 - Operations
 - Accounts
 - Vendor
-

8.2 Client Management Module

Purpose

Manage corporate client configurations.

Features

- Add/Edit Clients
- Configure billing cycles
- Configure import templates
- Manage pricing rules

Client Information

Field

Client Name

GST Number

Billing Cycle

Contact
Person

Contact Email

Active Status

8.3 Excel Upload & Import Module

Purpose

Allow uploading of client-specific transport Excel files.

Key Problem Solved

Different clients provide:

- different headers
- different structures
- different naming conventions

This module standardizes the import process.

Features

Upload Support

- .xlsx
- .csv

Import Flow

1. Upload file
 2. Select client
 3. Preview data
 4. Map columns
 5. Validate rows
 6. Process import
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Validation Rules

System should detect:

- duplicate trips
 - missing vehicle number
 - missing KM
 - invalid date
 - invalid vendor
 - invalid slab
-

8.4 Dynamic Column Mapping Engine

Purpose

Map uploaded Excel headers into standardized internal fields.

Example

Uploaded Header	Internal Field
FinalKMS	total_km
VehicleRegNo	vehicle_number
Boarded	employee_count

Features

- Manual mapping
 - Auto-suggestion
 - Save mapping templates
 - Reusable templates per client
-

8.5 Data Normalization Engine

Purpose

Convert heterogeneous client data into a unified trip schema.

Internal Standard Trip Schema

Field

trip_date
client_name
vendor_name
vehicle_number
vehicle_type
employee_count
total_km
bill_category
km_slab
trip_rate

Why This Matters

All downstream systems:

- settlements
- invoices
- reports
- dashboards

must use ONE internal structure.

8.6 Trip Management Module

Purpose

Store and manage normalized trips.

Features

- View trips
 - Edit trips
 - Filter trips
 - Search trips
 - Bulk update
-

Trip Status Flow

Draft

↓

Validated

↓

Processed

↓

Invoiced

↓

Settled

8.7 Pricing Engine

Purpose

Automatically calculate trip charges.

Supported Pricing Logic

KM Slab Pricing

Example:

Slab	Rate
0–15 KM	₹500
16–25 KM	₹700
26+ KM	₹900

Vehicle-Based Pricing

Vehicle Type	Rate Logic
4 Seater	Slab Based
6 Seater	Slab Based
EV	Separate Pricing

Future Pricing Rules

- waiting charges
 - night charges
 - surge pricing
 - hourly pricing
-

8.8 Vendor Settlement Module

Purpose

Generate monthly vendor payout summaries.

Core Logic

System groups:

- trips
- vehicle type
- KM slabs
- vendor

Then calculates:

- trip counts
 - total amount
 - deductions
 - payable amount
-

Settlement Example

Slab	Trips	Amount
0–15KM	5	₹2500
16–25KM	3	₹2100

Deductions Supported

- service charge
 - tax
 - advance
 - penalties
-

8.9 Invoice Generation Module

Purpose

Generate settlement invoices.

Features

- PDF invoice generation
 - downloadable invoices
 - invoice numbering
 - GST support
 - branding support
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Invoice Contents

Header

- Vendor Name
- Month
- Vehicle Number
- Vehicle Type

Settlement Breakdown

- slab-wise totals
 - trip counts
 - deductions
 - grand total
-

8.10 Dashboard & Analytics

Purpose

Provide operational and financial visibility.

Dashboard KPIs

KPI	Description
Total Trips	Completed trips
Total Revenue	Client billing
Vendor Cost	Settlement amount
Gross Margin	Revenue - Vendor Cost
Top Vendors	Highest trips
Client Revenue	Client-wise billing

8.11 Reports Module

Reports

Operational Reports

- Trip report
- Vendor report
- Vehicle utilization

Financial Reports

- Invoice report
- Settlement report
- Profit & Loss report

9. What We Need To Track

9.1 Trip Tracking

Data

Trip Date

Client

Vendor

Vehicle Number

Vehicle Type

KM

Employees
Count

Trip Category

Rate

Status

9.2 Financial Tracking

Data

Trip Revenue

Vendor Cost

Settlement Amount

Invoice Amount

Tax

Deductions

Net Payable

9.3 Vendor Tracking

Data

Vendor Trips

Vendor Revenue

Pending
Settlements

Paid Settlements

Vehicle
Performance

9.4 Operational Tracking

Data

Import Status

Validation Errors

Duplicate Trips

Invoice Status

Settlement
Status

10. What Should Be Visible To Clients

Client Portal (Future Scope)

Corporate clients should be able to view:

Billing Dashboard

Information

Total Trips

Monthly Billing

Vehicle Utilization

Shift Usage

Cost Summary

Invoice Access

Clients can:

- download invoices
 - view billing summaries
 - filter by month
-

Operational Reports

Clients can view:

- route summaries
 - vehicle type usage
 - KM summaries
 - shift-wise reports
-

11. Admin Dashboard Requirements

Operational Dashboard

Must Show

Live Metrics

- trips processed
 - pending imports
 - pending settlements
 - invoice generation status
-

Financial Metrics

- total revenue
 - vendor payout
 - margins
 - deductions
-

Alert Metrics

- failed imports
 - duplicate records
 - pricing mismatches
-

12. Key Product Differentiators

D1 — Multi-Format Import Engine

Supports heterogeneous Excel structures.

D2 — Dynamic Mapping System

Reusable mapping templates.

D3 — Settlement Automation

Vendor invoice generation automation.

D4 — Margin Visibility

Per trip/vendor/client profitability.

D5 — Standardized Internal Schema

Scalable enterprise architecture.

13. Technical Requirements

Recommended Stack

Frontend

- React
- TypeScript
- TailwindCSS

Backend

- Node.js (NestJS)

Database

- PostgreSQL

Queue

- Redis + BullMQ

Storage

- AWS S3
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14. Success Metrics

Metric	Goal
Billing Time	↓ 80%
Manual Work	↓ 90%
Settlement Errors	↓ significantly
Vendor Settlement Time	< 1 day
Import Processing Time	< 30 sec

15. Risks & Challenges

Risk	Mitigation
Different client formats	Dynamic mapping engine
Incorrect uploads	Validation engine
Pricing complexity	Configurable rule engine
Excel dependency	Familiar UI/exports
Vendor disputes	Audit & logs

16. Future Scope

Phase 2

- Driver management
 - Vehicle compliance
 - Client portal
 - WhatsApp notifications
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Phase 3

- GPS integrations
 - Mobile apps
 - API integrations
 - Approval workflows
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Phase 4

- AI anomaly detection
 - predictive cost analysis
 - smart settlement validation
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17. Final Product Principle

MOST IMPORTANT PRINCIPLE

No matter how different the uploaded client data is,
the system must convert all imported data into ONE standardized internal operational schema.

This architecture is the foundation of:

- scalability
- reporting

- settlements
- invoice generation
- analytics
- future enterprise growth.